

Technology Stack

Date: 30 September 2026 | Team ID: 6ab2240ffc67a5bc0222524ef | Project: ComicCraft – AI Comic Story Creator

Technology stack details

S.No	Architecture component	Technology	Purpose / justification
1	Frontend	HTML + CSS + Jinja2	Simple server-rendered UI for prompt entry and comic preview.
2	Backend	Python + FastAPI	Async-friendly web API, routing, validation integration and automatic API documentation.
3	Validation	Pydantic v2	Typed PromptRequest and ImageTestRequest models with field validation.
4	Text generation	Gemini model configuration + local/demo generators	The configuration supports Gemini 2.5 Flash/Pro; packaged generator modules currently provide deterministic fallback/demo generation.
5	Image generation	Hugging Face InferenceClient + FLUX.1-schnell	Produces panel illustrations from image prompts.
6	Image processing	Pillow	Supports generated image files and image handling dependencies.
7	PDF export	fpdf2	Creates downloadable multi-page comic PDFs.
8	Templates	Jinja2	Renders index, comic preview and export success pages.
9	Testing	pytest + httpx	Tests routes, layout assembly and required project files.
10	Server	Uvicorn	Runs the FastAPI application locally.

Configuration

Environment-sensitive values are represented in app/config.py and should be supplied through a local .env file. Secrets are not included in this submission package.